

Week 10 Formula Sheet: Transport Layer & Reliable Data Transfer

1. UDP Header Size

UDP header = 8 bytes (4 fields $\times$ 16 bits)

2. UDP Length

UDP Length = header + data

3. Transmission Time

$t_{tx} = L / B$

4. Stop-and-Wait Utilization

$U = t_{tx} / (RTT + t_{tx})$

5. Throughput

Throughput = $U \times B$

6. Bandwidth-Delay Product

$BDP = B \times RTT$

7. Window Size

$W = BDP / \text{packet size}$

8. Go-Back-N Constraint

$N \leq 2^k - 1$

9. Selective Repeat Constraint

$N \leq 2^{(k-1)}$

10. Key Insight

GBN retransmits full window; SR retransmits only lost packets